CAUSALITY ASSESSMENT

A concise professional summary of key concepts, assessment methods, and vaccine-related assessment

Purpose: To evaluate whether an adverse event may be causally associated with a medicine or vaccine, using clinical information and structured assessment methods.

1. Overview and Definition

Causality assessment, also called relatedness assessment, is the process of evaluating the possible causal relationship between exposure to a drug and an adverse event. The central question is whether the medicine may have caused the event. An event is generally described as an adverse drug reaction (ADR) when a causal relationship is considered to exist; where the relationship is unknown or not yet established, it remains an adverse event (AE).

The source material states that this assessment is primarily performed by a medical reviewer or safety physician in a pharmaceutical company. Conclusions depend on available evidence and clinical judgement.

2. Main Factors Considered in Causality Assessment

The source groups the assessment factors into four broad categories:

Factor category	Key considerations
Drug-related	Whether the reaction is already known for the medicine (labelledness/known association), predisposing factors, and concomitant medicines taken alongside the suspected drug.
Event-related	The nature and characteristics of the reported event and whether the event pattern is compatible with a drug-related reaction.
Patient-related	Medical and past history, allergies, predisposing factors, and differences in individual response, including possible pharmacogenomic influences.
Clinical	Timing of onset, what happens after stopping or restarting the medicine (dechallenge/rechallenge), and the response to treatment given to manage the event.

3. Important Clinical Concepts

Temporal relationship

This refers to the time between taking the medicine and the onset of the event. An onset that fits the known clinical pattern may support a relationship; timing that does not fit may suggest alternative explanations. A drug’s half-life may also be relevant.

Dechallenge and rechallenge

- **Dechallenge:** observe whether the event improves or resolves after the suspected medicine is stopped or its dose is reduced.
- **Rechallenge:** observe whether the event returns after the medicine is restarted. Reappearance may strengthen the suspected association.

Response to corrective treatment

The response to treatment for the adverse event may provide additional clinical information. However, response or non-response alone does not establish causality and must be interpreted with the rest of the case evidence.

Patient history and concomitant medicines

Previous allergies, earlier reactions to similar medicines, underlying conditions, and medicines used at the same time may affect the assessment. For example, a history of penicillin allergy may be relevant when evaluating a possible allergic reaction to a related medicine.

4. Common Causality Assessment Methods

The source identifies the Naranjo algorithm and the WHO–UMC causality assessment system as commonly used methods.

4.1 Naranjo Algorithm

The Naranjo algorithm uses 10 questions about the reported event and the suspected medicine. Responses receive predefined scores, which are added to produce a total. The score is then mapped to a causality category.

Total score	Category stated in the source
≥ **9**	Definite
5–8	Probable
1–4	Possible
0	Doubtful

The questions consider previous conclusive reports, whether the event followed drug administration, improvement after stopping or reducing the medicine, recurrence after re-administration, alternative causes, toxic drug concentrations, dose-response, prior similar reactions, and objective confirmation.

Illustrative example from the source

A patient, Raj, began amoxicillin 500 mg three times daily for a throat infection on 1 June 2021. Itchy rashes appeared the next day. The dose was reduced to 250 mg three times daily and the rash disappeared; after the dose was increased again, generalized itchy rashes reappeared. This sequence supports a possible causal relationship. The final Naranjo score depends on how each question is scored and on the clinical information available.

4.2 WHO–UMC Causality Assessment

The WHO–UMC system was developed by the World Health Organization and the Uppsala Monitoring Centre. It categorizes cases according to the strength of the evidence and the information available.

Category	General meaning
Certain	Evidence strongly supports a causal relationship.
Probable / Likely	The timing and clinical pattern support a relationship, with alternative explanations less convincing.
Possible	A relationship may exist, but alternative explanations or incomplete evidence remain.
Unlikely	The timing or clinical evidence makes a causal relationship less likely.
Conditional / Unclassified	More information is needed before a classification can be made.
Unassessable / Unclassifiable	Information is insufficient or contradictory and cannot be supplemented or verified adequately.

Unlike a purely numerical interpretation, WHO–UMC classification requires clinical judgement based on the available case information.

5. Implied Causality in Spontaneous Reports

A spontaneous (non-solicited) case report is one in which a patient, healthcare professional, or another reporter independently reports a suspected adverse event to a company or healthcare professional. The source emphasizes that a suspected causal relationship should not be dismissed simply because the company representative initially believes the event is unrelated. The report should be handled and evaluated under applicable pharmacovigilance procedures, because prematurely ruling out a relationship may affect appropriate safety reporting.

6. Key Terminology

Term	Meaning in the context of the source
Adverse event (AE)	An unfavourable occurrence after medicine use for which causality may be unknown or not established.
Adverse drug reaction (ADR)	An adverse event for which a causal relationship with the medicine is suspected or considered to exist.
Serious adverse event (SAE)	An adverse event that meets applicable seriousness criteria; seriousness is distinct from whether it was caused by the medicine.
SUSAR	Suspected Unexpected Serious Adverse Reaction: a suspected medicine-related reaction that is serious and unexpected.

Important distinction: “Serious” describes the regulatory seriousness of an event, while “causal” describes its relationship to a product. These concepts should not be treated as interchangeable.

7. Causality Assessment for AEFI

The source explains that adverse events following immunization (AEFI) are assessed using a vaccine-specific approach recommended by WHO, rather than simply applying the same process used for medicines.

Possible explanations for an AEFI

- **Vaccine-product-related reaction:** related to the vaccine itself or a product-quality issue.
- **Vaccine-quality or handling issue:** for example, a possible temperature-control problem during storage or transport, or contamination.
- **Immunization-process-related reaction:** associated with the vaccination process, including anxiety around vaccination.
- **Other event:** an event occurring after vaccination that may not be caused by the vaccine or the immunization process.

Five principles highlighted in the source

1. Strength of association between the event and vaccine administration.
2. Consistency of the reported association with other evidence or known patterns.
3. Specificity of the adverse event.
4. Temporal relationship between vaccination and event onset.
5. Biological plausibility: whether the vaccine’s characteristics could reasonably explain the event.

Four-step assessment process

1. Determine whether the event is eligible for AEFI causality assessment and gather relevant information using a checklist.
2. Review the case information and apply the assessment algorithm or flowchart, including whether a known causal relationship exists.
3. Evaluate the evidence and consider alternative explanations.
4. Classify the case and document the rationale. The source describes: A — consistent causal association to immunization; B — indeterminate; C — inconsistent causal association; or unclassifiable when information is

insufficient.

The source gives a high-level overview of the AEFI process rather than the full WHO checklist or decision algorithm; the complete procedure should be consulted when performing an actual assessment.

8. Automation and Machine Learning

The source describes growing interest in automating causality assessment. A machine-learning system could be trained on large collections of previously assessed cases to learn patterns in how reviewers classify events. A potential workflow is to enter case data, generate a suggested classification, and validate the system's performance.

- Automation may improve efficiency and support consistency.
- Training requires sufficiently large, reliable case datasets and clear reference assessments.
- Clinical review, validation, and appropriate oversight remain important; automation should not be assumed to replace expert judgement without evidence.

9. Key Takeaways

- Causality assessment evaluates whether a drug or vaccine may have caused a reported adverse event.
- The assessment considers drug, event, patient, and clinical factors together.
- Timing, dechallenge, rechallenge, medical history, concomitant medicines, and alternative causes are important evidence.
- The Naranjo algorithm uses 10 questions and a numerical score; WHO–UMC uses defined clinical categories.
- A spontaneous report should be evaluated appropriately rather than prematurely dismissed as unrelated.
- AE, ADR, SAE, and SUSAR describe different aspects of a safety case and should be distinguished carefully.
- AEFI assessment uses a vaccine-specific WHO approach, including review of product, quality, immunization-process, and other possible causes.
- Automation and machine learning may support assessment, but reliable data, validation, and expert oversight are essential.

Source Note

Prepared as a professional summary of the supplied video transcript on causality assessment in pharmacovigilance. This educational summary is not a substitute for current regulatory guidance, the complete Naranjo questionnaire, or official WHO–UMC and WHO AEFI assessment materials.